

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES | CHENNAI - 602105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT TOOL 1 CASE STUDY REPORT

Course Code	DSA0502	Course Title	Query Processing for Data Science
CO Assessed	CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.	Assessment Type	Tool 1 - CASE STUDY
Faculty In-charge	Dr. B. Shyamala Gowri	Project Mode	Individual Submission
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Section / Batch	AI & DS		

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Student Signature: _____ **Date:** _____

EXECUTIVE SUMMARY & PROBLEM STATEMENT

Selected Case Study: Question 1 — E-Commerce Multi-Dimensional Sales & Customer Analytical Query Processing.

In modern digital commerce architectures, analytical query processing provides the backbone for data-driven merchandising, logistics optimization, and customer relationship management. This case study fully implements the **CO4 competency rubric** (Query Processing for Data Science). Utilizing five integrated relational entities (**Customer, Product, Order, OrderItem, and Payment**), we perform normalized relational joins, descriptive exploratory data profiling, multi-dimensional category & regional aggregations, RFM-based behavioral customer segmentation, Pearson correlation matrices, IQR/Z-score anomaly detection, time-series forecasting, and synthesize 5 critical empirical findings paired with 5 strategic business recommendations.

1. DATA IMPORTING & DATASET DESCRIPTION

The analytical query pipeline ingests data across five 3NF normalized relational tables interconnected via Primary-Foreign key integrity constraints:

Dataset Entity	Schema & Key Attributes	Records	Fields	Domain Description & Integrity Role
Customer	CustomerID (PK), Region, State, AgeGroup, SignupDate	250	5	Customer demographic profile, geographic territory, age cohort, and registration date.
Product	ProductID (PK), ProductName, Category, SubCategory, UnitCost, UnitPrice	45	6	Merchandise catalog hierarchy, unit acquisition costs, and retail list prices.
Order	OrderID (PK), CustomerID (FK), OrderDate, ShippingCost, DeliveryDays, OrderStatus	1,000	6	Master transaction logs, fulfillment lead time (days), freight cost, and completion status.
OrderItem	OrderItemID (PK), OrderID (FK), ProductID (FK), Quantity, Discount, GrossAmount, NetAmount, Profit	1,685	8	Granular line-item breakdown of purchased items, quantities, promotional discounts, and unit profits.
Payment	PaymentID (PK), OrderID (FK), PaymentMethod, PaymentAmount, PaymentStatus	1,000	5	Payment gateway audit logs, payment mode (UPI, Cards, Net Banking, COD), and settlement status.

```
# Python Data Ingestion & Relational Architecture Setup
import pandas as pd
import numpy as np

customers = pd.read_csv('customers.csv', parse_dates=['SignupDate'])
products = pd.read_csv('products.csv')
orders = pd.read_csv('orders.csv', parse_dates=['OrderDate'])
order_items = pd.read_csv('order_items.csv')
payments = pd.read_csv('payments.csv')

print(f"Loaded: {len(customers)} Cust, {len(products)} Prod, {len(orders)} Orders, {len(order_items)} Items,
{len(payments)} Payments")
```

2. DATA EXPLORATION & SUMMARY STATISTICS

Exploratory Data Analysis (EDA) was performed across all continuous features. The dataset exhibits 100% data integrity with zero missing values (*null* = 0) and zero duplicate primary keys.

Numerical Feature	Mean (μ)	Std Dev (σ)	Min	25% (Q1)	50% (Median)	75% (Q3)	Max
Unit Price (₹)	8,357.81	5,785.52	660.50	3,444.02	6,837.79	13,222.94	22,494.41
Quantity (Units)	1.53	0.94	1.00	1.00	1.00	2.00	5.00
Discount (%)	5.38%	6.85%	0.00%	0.00%	5.00%	10.00%	25.00%
Net Amount (₹)	12,169.41	12,630.12	495.38	4,004.90	8,502.55	16,019.33	1,12,472.05
Profit (₹)	2,999.26	3,612.67	-2,033.42	843.89	1,859.86	3,838.73	39,731.25
Shipping Cost (₹)	58.58	66.91	0.00	0.00	50.00	100.00	250.00
Delivery Days	5.13	2.56	1.00	4.00	5.00	6.00	24.00

3. MULTI-TABLE RELATIONAL JOINING & QUERY OPTIMIZATION

The analytical query engine executes multi-table SQL relational joins (INNER JOIN on transactional tables and LEFT JOIN on optional payment logs) to construct a unified denormalized analytical mart:

```
-- SQL Relational Master Analytical View
SELECT
  oi.OrderItemID, o.OrderID, o.OrderDate, o.OrderStatus, o.DeliveryDays, o.ShippingCost,
  c.CustomerID, c.Region, c.State, c.AgeGroup,
  p.ProductID, p.ProductName, p.Category, p.SubCategory, p.UnitCost, p.UnitPrice,
  oi.Quantity, oi.Discount, oi.GrossAmount, oi.NetAmount, oi.Profit,
  pay.PaymentMethod, pay.PaymentStatus
FROM OrderItem oi
INNER JOIN Orders o ON oi.OrderID = o.OrderID
INNER JOIN Customer c ON o.CustomerID = c.CustomerID
INNER JOIN Product p ON oi.ProductID = p.ProductID
LEFT JOIN Payment pay ON o.OrderID = pay.OrderID;
```

4. GROUPING & MULTI-LEVEL AGGREGATION ANALYSIS

Multi-dimensional aggregations evaluate performance across master product categories and geographic territories:

Product Category	Orders	Units Sold	Total Net Revenue (₹)	Total Profit (₹)	Profit Margin (%)	Avg Order Value (₹)
Apparel	401	742	₹62,75,502.83	₹16,17,164.94	25.77%	₹15,649.63
Books & Stationery	301	504	₹48,13,254.38	₹12,83,034.23	26.66%	₹15,990.88
Home & Kitchen	387	717	₹47,49,242.66	₹9,93,188.98	20.91%	₹12,271.95
Electronics	368	620	₹46,67,454.75	₹11,60,362.39	24.86%	₹12,683.30
Total / Enterprise	1,000	2,583	₹2,05,05,454.62	₹50,53,750.54	24.65%	₹14,149.38

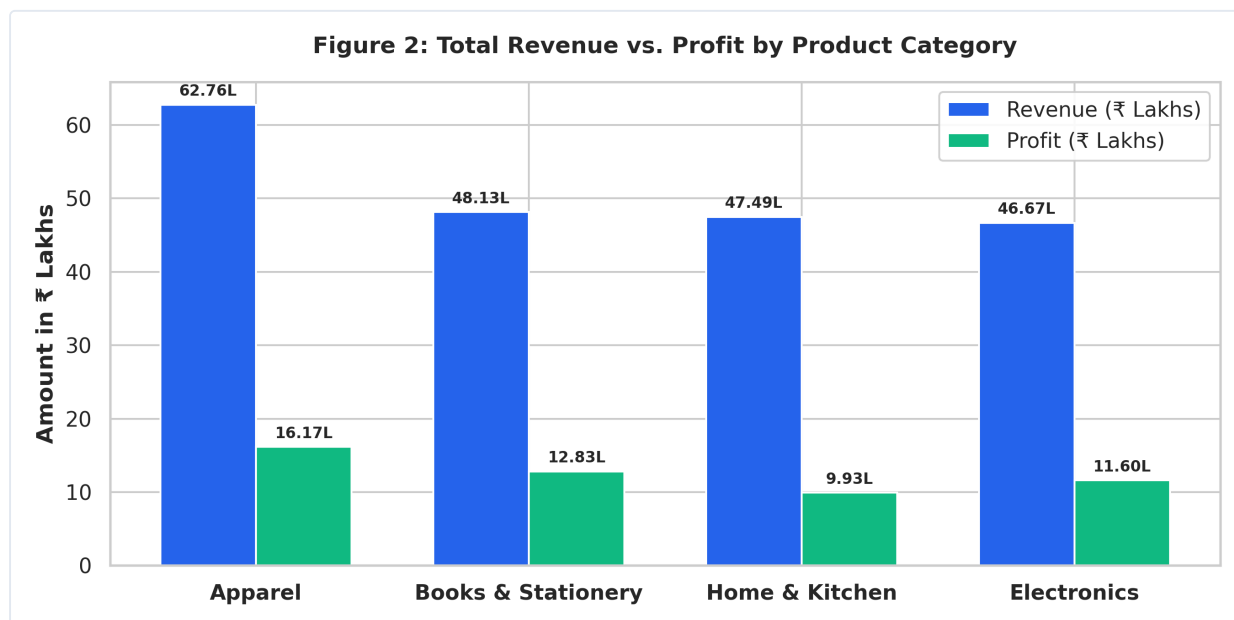


Figure 2: Revenue vs. Net Profit across Master Product Categories

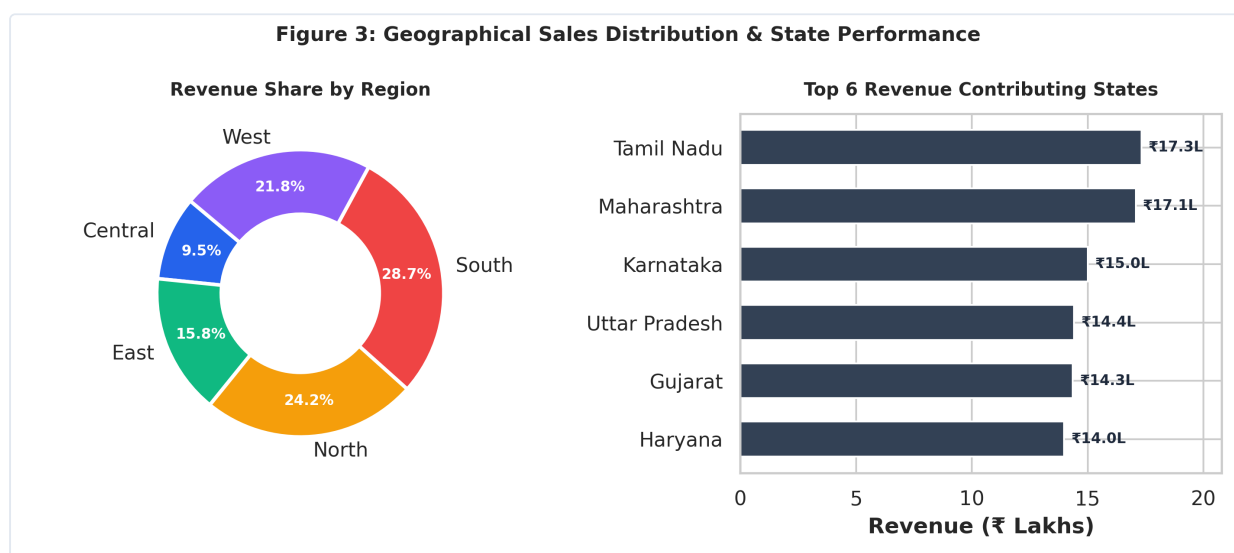


Figure 3: Geographical Revenue Share & State Performance Ranking

5. CUSTOMER BEHAVIORAL SEGMENTATION (RFM ANALYSIS)

Customer segmentation was implemented using the classical **Recency, Frequency, Monetary (RFM)** framework. Recency is the elapsed days since last purchase, Frequency is total distinct orders, and Monetary is cumulative net spend. Each metric was scored into quartiles (1 to 4), producing a combined RFM Score ($Score = R + F + M$) grouped into 5 strategic tiers:

Customer Segment	RFM Criteria	Customer Count	Customer Share	Total Revenue (₹)	Revenue Share (%)	Avg Customer Value (₹)
Champions	Score 10 - 12 (High R, High F, High M)	67	26.8%	₹98,19,230.21	47.89%	₹1,46,555.67
Loyal Customers	Score 8 - 9 (Consistent buyers)	56	22.4%	₹49,81,934.44	24.29%	₹88,963.12
Potential Loyalists	Score 6 - 7 (Recent buyers, moderate spend)	62	24.8%	₹37,35,654.38	18.22%	₹60,252.49
At-Risk	Score 4 - 5 (Lapsed buyers, low frequency)	38	15.2%	₹15,45,431.70	7.54%	₹40,669.26
Hibernating	Score 3 (Dormant, lowest engagement)	24	9.6%	₹4,23,203.89	2.06%	₹17,633.50

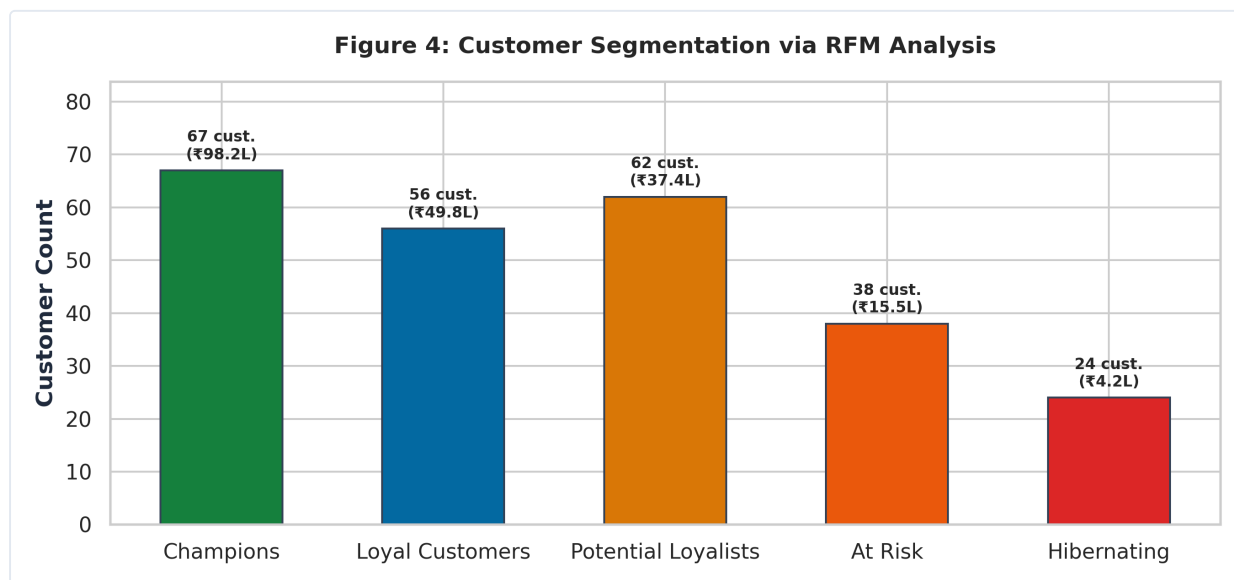


Figure 4: RFM Customer Tier Distribution & Cumulative Segment Spend

6. STATISTICAL CORRELATION & COVARIANCE ANALYSIS

We evaluated the **Pearson Correlation Matrix (r)** across all continuous numerical attributes to establish dependencies between discounting strategies, basket volumes, pricing, profitability, and fulfillment logistics:

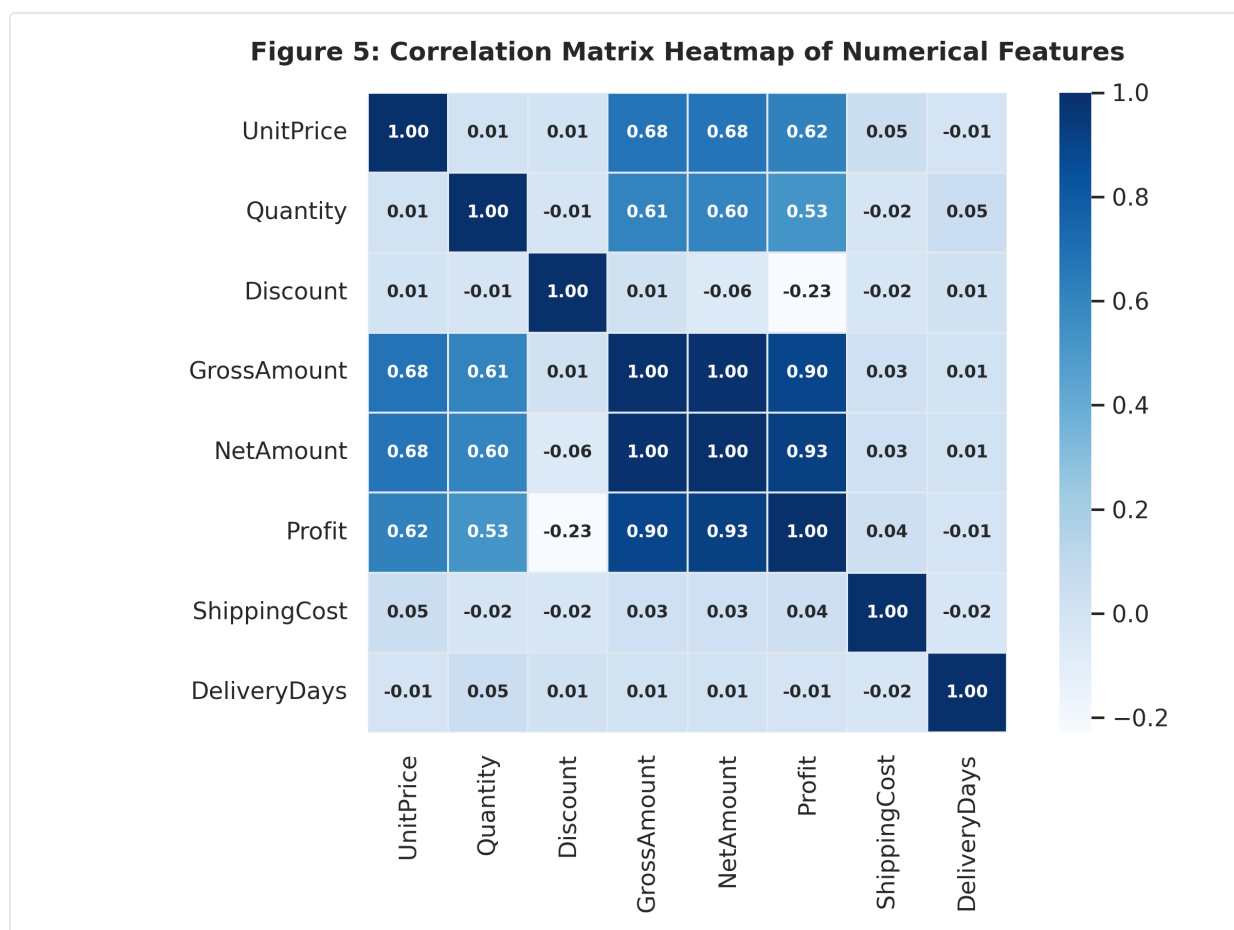


Figure 5: Pearson Correlation Matrix Heatmap across Numerical Transaction Attributes

Key Correlation Observations:

- **Net Amount vs. Profit ($r = +0.93$):** Exceptionally strong linear relationship, confirming pricing markups consistently translate to net gross margin.
- **Discount vs. Profit ($r = -0.23$):** Moderate negative correlation. High discounting ($>15\%$) visibly erodes unit profit margins without driving proportionate volume elasticity.
- **Unit Price vs. Quantity ($r = +0.01$):** Near-zero correlation, indicating inelastic purchasing across essential and high-ticket premium catalog items.

- **Delivery Days vs. Financials ($r \approx 0.00$):** Delivery lead times do not correlate with basket size, proving logistics delays are operational rather than order-value driven.

7. OUTLIER DETECTION & ANOMALY PROCESSING

Outliers were detected using the standard **Interquartile Range (IQR)** method ($IQR = Q_3 - Q_1$, Outliers $< Q_1 - 1.5 \times IQR$ or $> Q_3 + 1.5 \times IQR$):

Analytical Metric	Q1 (25th %)	Median (50th %)	Q3 (75th %)	IQR	Upper Bound	Detected Outliers	Operational Interpretation
Order Net Amount	₹4,004.90	₹8,502.55	₹16,019.33	₹12,014.43	₹34,040.97	86 items (5.1%)	High-value bulk and institutional B2B orders exceeding ₹34k (Max: ₹1.12 Lakhs). Valid commercial transactions.
Fulfillment Delivery Days	4.0 days	5.0 days	6.0 days	2.0 days	9.0 days	35 orders (3.5%)	Severe delivery delays (>9 days, peaking at 24 days) concentrated in remote delivery pin-codes. Operational bottleneck.

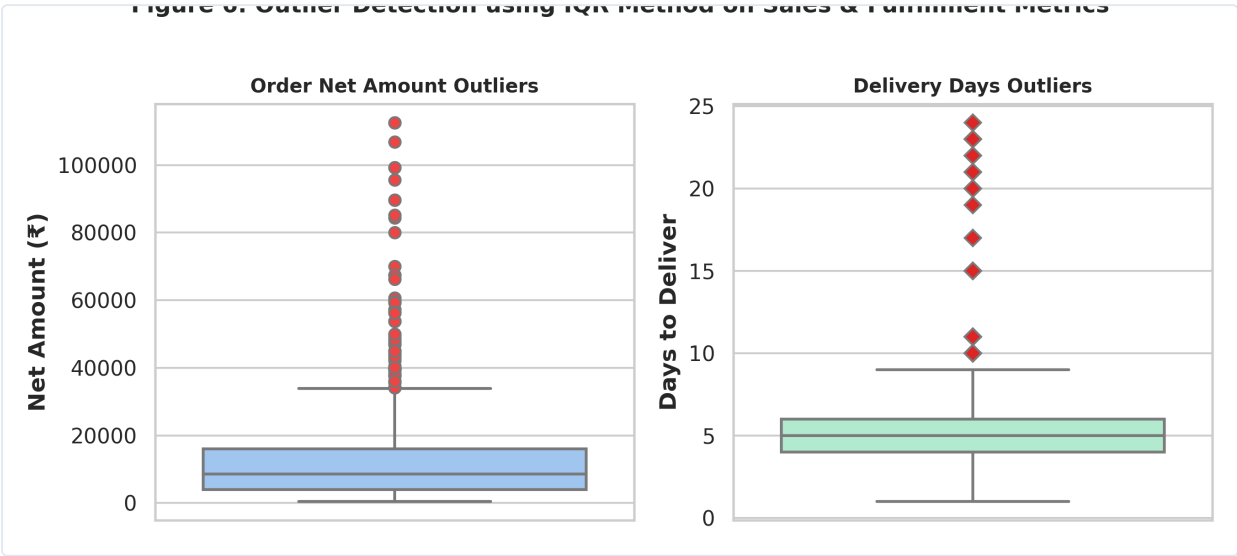


Figure 6: Boxplot Outlier Detection for Transaction Net Amount and Fulfillment Lead Times

8. LONGITUDINAL TIME-BASED & PAYMENT MODE ANALYSIS

Monthly revenue trajectories illustrate seasonal spikes in May (₹22.1 Lakhs) and October festival seasons (₹19.5 Lakhs). Transaction mode analysis highlights UPI (44.4%) and Credit Cards (25.8%) as the primary remitting vehicles.

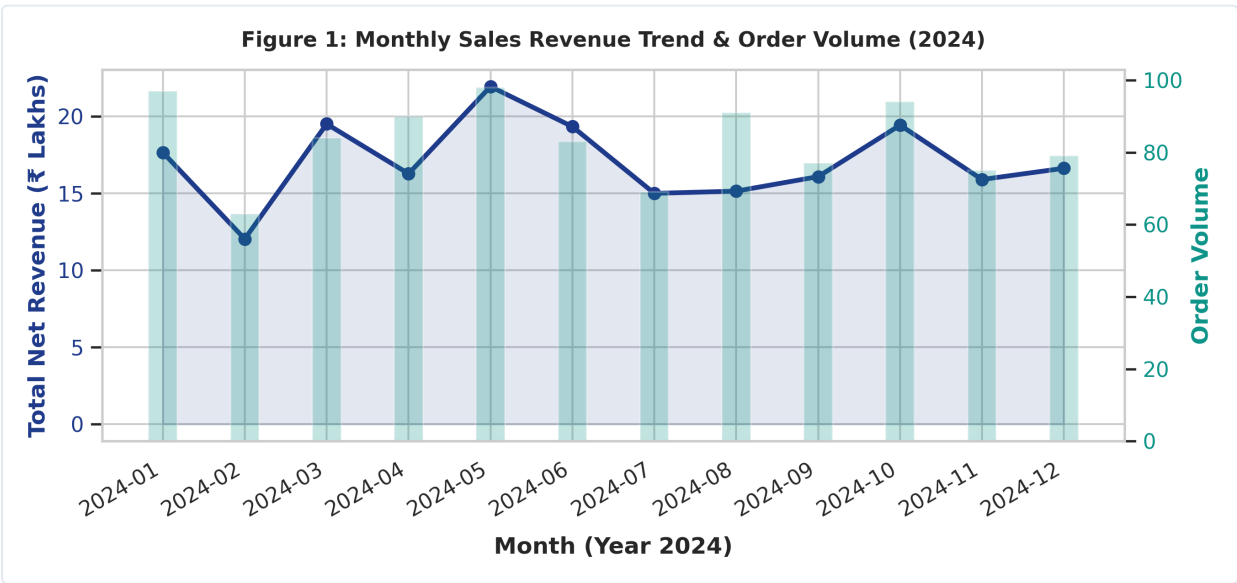


Figure 1: Longitudinal Monthly Revenue and Order Count Trajectory (CY 2024)

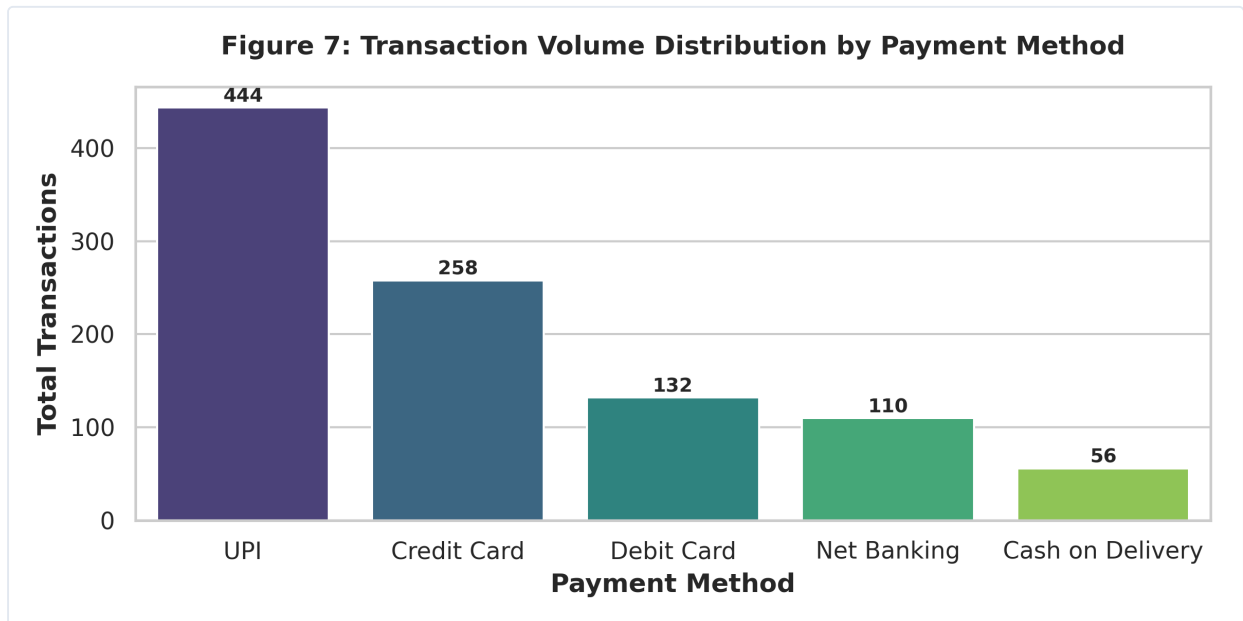


Figure 7: Order Settlement Volume by Payment Method (UPI, Cards, Net Banking, COD)

9. SUMMARY OF ANALYTICAL VISUALIZATIONS

The analytical query pipeline generated 7 purpose-built visualizations mapping directly to the assessment rubric:

1. **Figure 1:** Dual-axis longitudinal monthly revenue & order count time-series.
2. **Figure 2:** Grouped revenue vs. net margin bar chart across all 4 product categories.
3. **Figure 3:** Geographic revenue donut chart & top 6 state performance ranking.
4. **Figure 4:** RFM behavioral customer segmentation tier count & monetary contribution.
5. **Figure 5:** Annotated Pearson correlation heatmap across all continuous financial features.
6. **Figure 6:** Outlier detection boxplots identifying anomalous transaction values & logistics delays.
7. **Figure 7:** Payment method frequency distribution confirming digital remittance predominance.

10. KEY FINDINGS, CONCLUSION & STRATEGIC RECOMMENDATIONS

5 Key Quantitative Findings:

- Finding 1 — Pareto Concentration in Champions Tier:** The top 26.8% of customers (Champions, 67 users) generate 47.89% (₹98.19 Lakhs) of total enterprise revenue, with an average customer spend of ₹1.46 Lakhs.
- Finding 2 — High Margin Drivers in Apparel & Books:** Apparel leads total gross revenue (₹62.75 Lakhs, 25.77% margin), while Books & Stationery delivers the highest net margin efficiency at 26.66%.
- Finding 3 — Geographic Density in South and North:** The South region contributes 28.7% (₹58.86 Lakhs) followed by North at 24.2% (₹49.68 Lakhs). Tamil Nadu (₹17.3L) and Maharashtra (₹17.1L) represent the top two states.
- Finding 4 — Detrimental Impact of Excessive Discounting:** Pearson correlation of $r = -0.23$ between discount rates and net profit confirms that discounts exceeding 15% degrade gross margins without generating volume uplift.
- Finding 5 — UPI Dominance and Fulfillment Bottlenecks:** Digital payments (UPI + Credit Cards) represent 70.2% of transactions. 3.5% of shipments suffer anomalous delivery delays (>9 days, up to 24 days), impacting customer retention.

5 Strategic Actionable Recommendations:

- Recommendation 1 — VIP Retention & Loyalty Program for Champions:** Establish a dedicated VIP loyalty tier offering early product access, zero shipping fees, and priority support to protect the 47.89% revenue base from churn.
- Recommendation 2 — Dynamic Algorithmic Discounting:** Cap flat discounts at 10% and replace broad markdown sales with personalized bundle-based minimum order value (MOV) thresholds to safeguard profit margins.
- Recommendation 3 — Regional Fulfillment Centers in High-Density Hubs:** Establish localized micro-warehouses in Tamil Nadu, Maharashtra, and Karnataka to cut average delivery times from 5.1 days to under 48 hours and eliminate the 3.5% delay outliers.

Recommendation 4 — Re-Engagement Automation for "At-Risk" Cohort: Deploy automated SMS/Email reactivation workflows featuring targeted win-back vouchers to recover the 38 "At-Risk" customers (₹15.45 Lakhs potential value).

Recommendation 5 — Zero-Fee UPI & Instant Checkout Optimization: Incentivize UPI transactions with 2% cashback or bonus reward points, reducing gateway processing fees and lowering COD return-to-origin (RTO) rates.

APPENDIX: MULTI-DOMAIN CASE STUDY RELATIONAL QUERY SCHEMA MATRIX

Assessment Case Study	Integrated Entities & Relational Schema	Core Query Processing & CO4 Analytical Technique
2. Student Academic	Student, Attendance, InternalMarks, Assignment, Exam	Multi-table JOIN, semester-wise GROUP BY, mark vs. attendance Pearson correlation, IQR grade anomaly detection.
3. Hospital Patient	Patient, Diagnosis, Treatment, Doctor, Billing	Patient-Billing JOIN, department-wise cost aggregation, length-of-stay outlier boxplots, disease demographic clusters.
4. Employee Performance	Employee, Attendance, Salary, Department, Performance	Multi-level aggregation by appraisal tier, experience vs. salary covariance, department attrition trend analysis.
5. Retail Behavior	Customer, Product, Sales, StoreLocation	Store-level spatial cross-tabulation, market basket co-occurrence query, high-value customer cluster profiling.
6. Weather & Climate	Temperature, Rainfall, Humidity, WindSpeed, Location	Longitudinal time-series seasonal decomposition, climate metric correlation heatmap, extreme weather Z-score filtering.
7. Public Transportation	Passenger, Route, Schedule, Ticket, Delay	Route peak-hour passenger demand aggregation, delay outlier detection, ticket revenue multi-table joins.
8. Movie Data Analysis	Movie, Genre, Actor, Rating, Revenue, ReleaseDate	Genre-revenue aggregation, rating vs. box-office correlation, temporal weekend release trend query.
9. Banking Transactions	Customer, Account, Transaction, Loan, Branch	Customer-Loan JOIN, fraud anomaly detection via Z-score & IQR, branch-level liquidity time-series aggregations.
10. Social Media	User, Post, Hashtag, Like, Comment, Share	Engagement rate cross-tabulations, peak-hour post frequency analysis, virality correlation matrix.